

such as wireless communication, self-charging or even self-healing. The next step is taking the technology of stretchable electronics from the laboratory to the market. This requires cheaper materials and faster, scalable manufacturing methods.

Further course of technology

Understanding of nano-scale phenomena, functional materials and soft devices has progressed to a point where substantial advances in stretchable electronics applications are becoming realistic. Materials, in general, when subjected to repeated stretch and restore process, tend to degrade in mechanical strength and recovery performance. **Electronic devices worn on the body or embedded into clothing have to be made from materials that can withstand repeated use and allow no change in the mechanical properties of the device even after repeated stretching and restoration.**

Panasonic Corp. has developed a polymer resin film enabling the construction of soft, flexible and stretchable electronic devices such as those developed for clothing or worn on the body. It is deployable in a broad range of applications, from wearable devices to sensors, displays and robots. This newly-developed material is an insulating film material that is said to stretch and return to its original shape—a feature that is hard to find in conventional flexible materials. It adapts to desired manners of folding and to varying free-form surfaces, substantially reducing existing design constraints. The company is also said to provide a transparent electrode material and conductive paste along with this insulating film.

Engineers at University of California San Diego, USA, have developed the first printed battery using zinc along with polymer material made from isoprene, making the battery flexible, stretchable and rechargeable. This stretchable battery can power every stretchable electronic device, from wearable sensors to solar cells and other kinds of electronics. The substance allows the battery to stretch to twice its size, in any direction, without suffering damage.

Future vision is to make 3D stretchable electronics that are as multifunctional and high performing as today's rigid electronics. Another problem while building 3D stretchable electronics is creating electrical connections between the layers. These electrical connections, known as vertical interconnect accesses (VIAs), are essentially small conductive holes that go through different layers on a circuit. The VIAs are traditionally made using lithography and etching techniques, which work fine on rigid electronics but are questionable for their suitability on stretchable elastomers.

Presently, researchers are mixing silicone elastomer with a black organic dye so that it could absorb energy from a laser beam, forming circuits onto each layer of the elastomer. Afterwards, they are looking forward to fill in the VIAs with conductive materials to electrically connect the layers to one another. It is being reported that full rubbery integrated electronics can further be obtained by introducing metallic carbon nanotubes into a rubbery semiconductor composite from a rubbery semiconductor. **EFY**

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